Machine Learning Internship Task: Customer Churn Prediction System

Objective

Develop an end-to-end machine learning application that predicts whether a customer is likely to leave a subscription service.

1. Data Collection

Use a publicly available dataset (e.g., Telco Customer Churn from Kaggle). Perform exploratory data analysis (EDA). Document all observations.

2. Data Preprocessing

Handle missing values, remove duplicate records, encode categorical variables, normalize/standardize numerical features where appropriate, and split the dataset into training and testing sets.

3. Feature Engineering

Perform feature selection, explain feature importance, and compare model performance before and after feature engineering.

4. Model Development

Train and compare Logistic Regression, Random Forest, and XGBoost (or LightGBM). Evaluate using Accuracy, Precision, Recall, F1-score, and ROC-AUC.

5. Hyperparameter Tuning

Use GridSearchCV or RandomizedSearchCV and report performance improvements.

6. Model Explainability

Use SHAP or LIME to explain feature importance and individual predictions.

7. REST API

Develop a Flask or FastAPI API exposing POST /predict that accepts customer details and returns a prediction with probability.

8. Web Interface

Create a Bootstrap-based HTML interface for user input and prediction display.

9. Model Persistence

Save the trained model using Pickle or Joblib and load it in the API without retraining.

10. Documentation

Include installation steps, dataset information, training procedure, API documentation, screenshots, and future improvements.

Bonus Tasks

Dockerize the application, add unit tests, and log predictions to SQLite or PostgreSQL.

Suggested Folder Structure

Customer-Churn-Prediction/
■■■ data/
■■■ notebooks/
■■■ models/
■■■ api/
■■■ templates/
■■■ static/
■■■ utils/
■■■ requirements.txt
■■■ train.py
■■■ predict.py
■■■ README.md
■■■ report.pdf